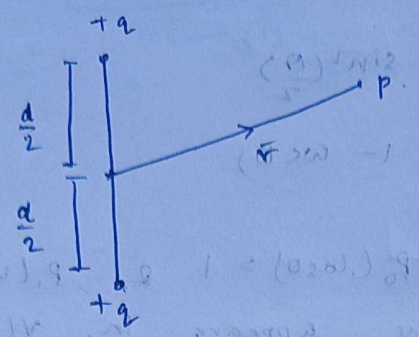
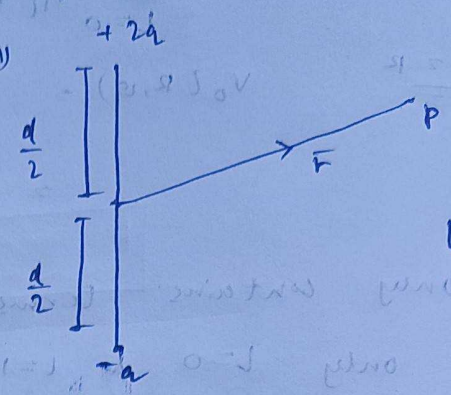
# Assignment - 4

① Compute the monopole & dipole moments of the following charge distributions.

(i)




(ii)




[ $r \gg d$ ]

Ans: Monopole : $Q = \int d^3r' \, \rho(\bar{r}') =$ Total charge of configurat$^n$.

Dipole : $\bar{p} = \int d^3\bar{r}' \, \rho(\bar{r}') \, \bar{r}'$

$=$ ~~the~~ $\sum$ charge $\times$ displacement vector for all charges

Configurat$^n$ 1 :

$Q_1 = 2q$

$\bar{p}_1 = q\left(\frac{\bar{d}}{2}\right) + q\left(-\frac{\bar{d}}{2}\right) = 0$

Configurat$^n$ 2 :

$Q_2 = q$

$\bar{p}_2 = 2q\left(\frac{\bar{d}}{2}\right) + (-q)\left(-\frac{\bar{d}}{2}\right) = q\bar{d} + \frac{q\bar{d}}{2} = \frac{3q\bar{d}}{2}$

Alternative :

(I) $\rho_1(\bar{r}') = q\,\delta^3\left(\bar{r}' - \frac{\bar{d}}{2}\right) + q\,\delta^3\left(\bar{r}' + \frac{\bar{d}}{2}\right)$

(II) $\rho_2(\bar{r}') = 2q\,\delta^3\left(\bar{r}' - \frac{\bar{d}}{2}\right) - q\,\delta^3\left(\bar{r}' + \frac{\bar{d}}{2}\right)$

$\therefore Q_1 = \int d^3\bar{r}' \, \rho_1(\bar{r}') = \int d^3r' \left[ q\,\delta^3\left(\bar{r}' - \frac{\bar{d}}{2}\right) + q\,\delta^3\left(\bar{r}' + \frac{\bar{d}}{2}\right)\right]$

$= q + q = 2q$

$\therefore \bar{p}_1 = \int d^3r' \, \rho_1(\bar{r}') \, \bar{r}'$

$= \int d^3r' \left[ q\,\delta^3\left(\bar{r}' - \frac{\bar{d}}{2}\right)\bar{r}' + q\,\delta^3\left(\bar{r}' + \frac{\bar{d}}{2}\right)\bar{r}'\right] = q\frac{\bar{d}}{2} - q\frac{\bar{d}}{2}$

$= 0$

$$\therefore \rho_2 = \int d^3r' \, \rho_2(\bar{r}')$$

$$= \int d^3r' \left[ 2q\,\delta^3\left(\bar{r}' - \frac{\bar{d}}{2}\right) - q\,\delta^3\left(\bar{r}' + \frac{\bar{d}}{2}\right) \right]$$

$$= 2q - q = q.$$

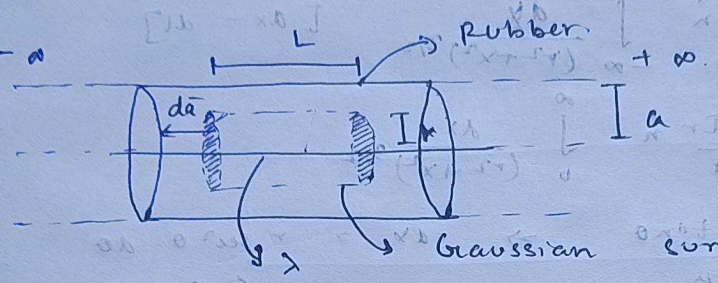
$$\therefore \bar{P}_2 = \int d^3r' \, \rho_2(\bar{r}')\, \bar{r}'$$

$$= \int d^3r' \left[ 2q\,\delta^3\left(\bar{r}' - \frac{\bar{d}}{2}\right)\bar{r}' - q\,\delta^3\left(\bar{r}' + \frac{\bar{d}}{2}\right)\bar{r}' \right]$$

$$= 2q\frac{\bar{d}}{2} - q\left(-\frac{\bar{d}}{2}\right) = q\frac{\bar{d}}{2} + q\frac{\bar{d}}{2} = \frac{3q\bar{d}}{2}$$

(2) A very long straight thin wire is carrying a uniform charge density $\lambda$ & is insulated by rubber of radius $a$. Find $\bar{D}(\bar{r})$ inside rubber $(r < a)$.

Ans:




Let us consider the Gaussian surface as shown.

$$\oint_G \bar{D}\cdot d\bar{a} = \underbrace{\int \bar{D}\cdot d\bar{a}}_{\text{(flat surfaces)}} + \underbrace{\int \bar{D}\cdot d\bar{a}}_{\text{(lateral surfaces)}}$$

$[\bar{D} \perp d\bar{a}]$

$$= D\int da \qquad [d\bar{a} = da\,\hat{r}, \quad \bar{D} = D\hat{r} \text{ (symmetrics)}]$$

$$= D(2\pi r L)$$

From Gauss's law:

$$\oint \bar{D}\cdot d\bar{a} = q_f \Rightarrow D(2\pi r L) = \lambda L$$

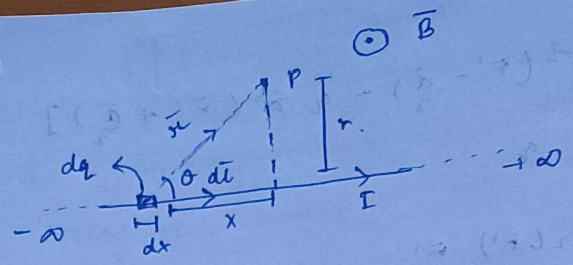
$$\Rightarrow D = \frac{\lambda}{2\pi r}$$

$$\therefore \boxed{\bar{D}(\bar{r}) = \frac{\lambda}{2\pi r}\hat{r}} \quad \text{(Ans)}$$

(3) Find the magnetic field @ a distance $r$ from a long wire carrying a steady current $I$ using:

(a) Biot-Savart's Law (b) Ampere's Law

Ans: (a)




According to Biot - Savart's law:

$$d\bar{B} = \frac{\mu_0}{4\pi} \int I \frac{d\bar{l} \times \hat{r}}{r^2}$$

$r^2 = r^2 + x^2$

$d\bar{l} \times \hat{r} = dl \sin\theta \, \hat{n} = \frac{dl\, r}{r} \hat{n}$ [$\hat{n}$ - perpendicular vector out of the plane of the paper]

$$\therefore d\bar{B} = \frac{\mu_0}{4\pi} \int \frac{I \, dl \, r \, \hat{n}}{(r^2+x^2)^{3/2}}$$

$$= \frac{\mu_0 I r \hat{n}}{4\pi} \int_{-\infty}^{\infty} \frac{dx}{(r^2+x^2)^{3/2}} \quad [dx = dl]$$

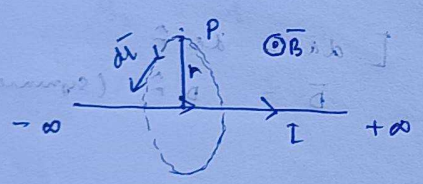
$$= \frac{\mu_0 I r \hat{n}}{2\pi} \int_0^{\infty} \frac{dx}{(r^2+x^2)^{3/2}}$$

Let $x = r \tan\theta \Rightarrow dx = r \sec^2\theta \, d\theta$

$$\therefore \int_0^{\infty} \frac{dx}{(r^2+x^2)^{3/2}} = \int_0^{\pi/2} \frac{\cancel{r} \sec^2\theta \, d\theta}{r^3 \cancel{\sec^3\theta} \sec\theta} = \frac{1}{r^2} \sin\theta \Big|_0^{\pi/2} = \frac{1}{r^2}$$

$$\therefore \bar{B} = \frac{\mu_0 I}{2\pi} \cdot \frac{1}{r^2} \cdot \hat{n} = \boxed{\frac{\mu_0 I}{2\pi r} \hat{n} = \bar{B}} \text{ (Ans)}$$

(b)




Consider an Amperian loop as shown in the figure.

Due to symmetry, $|\bar{B}|$ in the loop must be the same @ all points.

$$\oint_{loop} \bar{B} \cdot d\bar{l} = \mu_0 I \quad \text{(Ampere's law)}$$

$\Rightarrow \oint B \, dl = \mu_0 I$ [$\bar{B}$ & $d\bar{l}$ always point in the same directn along the loop. - magnetic field lines are closed loops around the wire]

$\Rightarrow B \oint dl = \mu_0 I$

$\Rightarrow B = \frac{\mu_0 I}{2\pi r}$

$$\boxed{\bar{B} = \frac{\mu_0 I}{2\pi r} \hat{n}} \text{ (Ans)}$$